Software Project Management

M2 — Project Plan Baseline Report

Project title: GoodTracker - Mobile Equipment Reservation Application

Team name / number: GoodTracker

Team members and roles: Talha Yılmaz - mobile UI; Eylül Melda Çakmak - mobile integration and testing; Kerem Tuna - database and equipment API; Kardelen Dargın - reservation API and project tracking

Submission date: 1 November 2026

Document version: 1.0 — working baseline adopted for planning on 1 November 2026

Repository / project board link: https://github.com/keremtuna0/good_tracker | https://github.com/users/keremtuna0/projects/2

1. Executive Summary

This M2 report sets the working plan for GoodTracker, a small Android demo for reserving shared laboratory and office equipment. The backlog contains eight Must, one Should and one Could work package. The Must scope is sign-in, equipment browsing and search, availability, reservation requests and cancellation, staff approval, and prevention of overlapping bookings. Self-registration, public borrower identity, payments, university SSO, push notifications and store release are outside the first demo.

The team uses the M1 hybrid approach: fixed academic milestone reporting with iterative development. The working forecast is 141.5 person-hours across eight delivery weeks, 2 November–27 December 2026. Each member can contribute 5–6 hours weekly, as reported by Kerem on 1 November. The estimate uses optimistic, most-likely and pessimistic hours for each item. The proposed allocation is Kerem 35.7 h, Kardelen 39.7 h, Talha 37.8 h and Eylül 28.3 h; unused capacity is reserve for exams, defects and changes. The official M3–M5 DYS dates must replace internal target dates when available.

As of 1 November, the M1 report and M2 planning source files exist, but the mobile and backend application have not been implemented and no application tests have run. M2 product issues #33–#39 now link to the backlog. The scope, estimates, schedule, RACI, communication plan and risk register form the working baseline described here. This report records planned work, not delivered features. DYS submission is a separate final step.

2. Scope Baseline

ID	Epic / Work package	Description	Priority	Acceptance criterion / Deliverable
TSK-01	Stakeholders and map	M1 stakeholder analysis and assumptions	Must	M1 stakeholder document and grid; user needs remain assumptions until checked.
TSK-02	Product backlog	IDs, priorities,	Must	Backlog covers

		dependencies and issue links		product and management work with links.
TSK-03	Sign-in and roles	Demo accounts and borrower/staff permissions	Must	Sign-in works; borrower requests cannot perform staff-only actions.
TSK-04	Equipment management	Staff create/update items and status	Must	Staff can manage items; borrowers can view them.
TSK-05	Conflict-free booking	Time-slot requests and overlap control	Must	At most one simultaneous overlapping request is accepted.
TSK-06	Equipment status view	Availability and return timing without public borrower identity	Should	Regular users see availability and expected return timing.
TSK-07	In-app reminder	Reminder for the signed-in borrower; no push	Could	If implemented, reminder appears at the agreed threshold.
TSK-08	Reports and presentation	Shared milestone reports and demo narrative	Must	Reports, evidence and reproducible demo are ready at their milestones.
TSK-09	Cancellation and staff approval	Own-request cancellation; staff approve/reject	Must	Borrowers cannot change others' bookings or use staff actions.
TSK-10	Equipment search	Category/location filters with availability	Must	Matching sample items and availability appear.

2.2 Scope Boundaries

The reservation-focused Android demo remains the M1 objective. The M2 backlog clarifies the boundary between Must, Should and Could work. These are the changes from the earlier backlog and M1 wording:

1. TSK-09 makes cancellation and staff approval explicit; both were already in the M1 scope.

2. TSK-10 makes category/location search explicit and Must, because M1 included it in the first demo.

3. TSK-07 is Could and limited to an in-app reminder; push notifications remain excluded.

4. TSK-03 excludes self-registration, TSK-06 excludes public borrower identity, and TSK-05 explicitly requires overlap prevention. These clarify the first-demo boundary.

3.1 Estimation Method

We use three-point PERT estimation in person-hours: E = (optimistic + 4 × most likely + pessimistic) / 6. The first eight TSK figures came from the team's M2 draft; six supporting-work estimates were added so prototype, data-model, mobile UI, integration, acceptance checks and weekly coordination are not silently omitted. These are planning

judgments, not measured actuals. The companion workbook gives each item's range, assumption and owner; actual hours will be recorded weekly and compared in M3. Calendar windows in the schedule are separate from effort.

3.2 Estimates

ID	Work item	Size / Effort estimate	Estimation method	Assumptions	Owner
TSK-01	Stakeholder analysis	Completed in M1; actual hours not recorded	n/a	M1 evidence only	Talha
TSK-02	Backlog drafting	Completed in M2; actual hours not recorded	n/a	Remaining issue links completed	Talha
TSK-03	Authentication and roles	O 8 / M 12 / P 20; E 12.7 h	Three-point PERT	Supabase auth; borrower and staff roles	Kerem
TSK-04	Equipment registration and status	O 6 / M 10 / P 16; E 10.3 h	Three-point PERT	Simple equipment fields and three statuses	Kerem
TSK-05	Conflict-free time-slot booking	O 10 / M 16 / P 28; E 17.0 h	Three-point PERT	One accepted overlapping booking per item	Kardelen
TSK-06	Equipment status and return timing	O 5 / M 8 / P 14; E 8.5 h	Three-point PERT	No public borrower identity	Talha
TSK-07	In-app return reminder	O 3 / M 5 / P 9; E 5.3 h	Three-point PERT	No push notifications	Eylül
TSK-08	Final reports and presentation	O 8 / M 12 / P 18; E 12.3 h	Three-point PERT	M3-M5 reports and final presentation	Kardelen
TSK-09	Cancellation and staff approval	O 6 / M 10 / P 16; E 10.3 h	Three-point PERT	Uses TSK-05 reservation model	Kardelen
TSK-10	Equipment search	O 4 / M 6 / P 10; E 6.3 h	Three-point PERT	Category and location filters	Talha
SUP-01	Technology prototype	O 4 / M 6 / P 10; E 6.3 h	Three-point PERT	Minimal Expo/API/Supabase connection	Kerem
SUP-02	Data model and API contract	O 4 / M 6 / P 10; E 6.3 h	Three-point PERT	Equipment, roles and reservations; shared API shapes	Kerem
SUP-03	Mobile navigation and browsing UI	O 8 / M 12 / P 20; E 12.7 h	Three-point PERT	Android demo screens with sample data	Talha
SUP-04	Mobile/API integration	O 8 / M 12 / P 20; E 12.7 h	Three-point PERT	Agreed API contract and working endpoints	Eylül
SUP-05	Acceptance checks and defect review	O 6 / M 10 / P 16; E 10.3 h	Three-point PERT	Named build and test data available	Eylül
SUP-06	Weekly tracking and decisions	O 6 / M 10 / P 16; E 10.3 h	Three-point PERT	Talha coordinates weekly evidence; Kardelen reviews milestone report	Talha
TOTAL	Future product and support work	141.5 person-hours	Three-point PERT	8-week working allocation; revise against actuals	All

3.3 Assumptions Log

ID	Assumption	Impact if invalid	Validation action	Owner
A-01	Users need equipment availability and reservation requests.	Product flow may change.	Observe prototype feedback.	Talha
A-02	Staff maintain equipment and approve requests.	Role model may change.	Confirm with a relevant staff member.	Talha
A-03	Expo, Express and Supabase support the demo.	Stack or scope may change.	Build a small prototype.	Kerem
A-04	Sample equipment data is available.	Demo preparation may slip.	Prepare and review sample data.	Kerem
A-05	Three-point figures are useful starting estimates.	Work may take longer.	Track actuals and reforecast.	Kerem
A-06	Each person can spend 5–6 h/week.	Dates or workload may slip.	Check capacity each week, especially around exams.	Kardelen
A-07	Work-package boundaries avoid double counting.	Total effort may be overstated.	Review item boundaries as implementation starts.	Kerem
A-08	Eight delivery weeks are available.	Official dates may require replanning.	Check M3–M5 DYS dates when published.	Kardelen

4. Schedule / Release Plan

Milestone or sprint	Goal	Start date	End date	Key dependencies	Exit criteria
M2 baseline	Finish and submit the shared planning report	26 Oct 2026	1 Nov 2026	M1 documents, M2 source files	DYS submission confirmed.
Sprint 1	Prototype, sign-in and first data-model version	2 Nov 2026	15 Nov 2026	M2 baseline and sample data	Prototype findings and named sign-in build.
Sprint 2	Equipment API, browsing/search and reservation increment	16 Nov 2026	29 Nov 2026	Data-model draft and API contract	Named increment with linked commits; defects recorded.
Sprint 3	Reservation lifecycle and mobile/API integration	30 Nov 2026	13 Dec 2026	Equipment API and staged reservation endpoints	Integrated flow on a named build.
Sprint 4	Conflict checks,	14 Dec 2026	27 Dec 2026	Integrated flow	No overlapping

	acceptance evidence and Android demo			and sample data	accepted bookings; reproducible demo.
M3–M5 course gates	Submit the required milestone reports	DYS date pending	DYS date pending	Official assignment pages	DYS receipts; internal sprint dates adjusted if needed.

The sprint dates are internal working targets, not official course deadlines. M3–M5 DYS dates were not verified on 1 November. Kardelen will update this schedule when those assignment pages become available; scope or capacity changes use the Section 7 change rule.

5.1 RACI Matrix

Activity	Responsible	Accountable	Consulted	Informed
Backlog and scope	Talha	Kardelen	Kerem, Eylül	Whole team
Effort estimates	Kerem	Kerem	Talha, Eylül, Kardelen	Whole team
Schedule and milestone report	Kardelen	Kardelen	Kerem, Talha, Eylül	Whole team
Mobile UI and search	Talha	Talha	Eylül, Kerem	Kardelen
Auth, data and equipment API	Kerem	Kerem	Kardelen, Eylül	Talha
Reservation API	Kardelen	Kardelen	Kerem, Eylül	Talha
Mobile/API integration	Eylül	Eylül	Talha, Kerem, Kardelen	Whole team
Acceptance checks	Eylül	Eylül	Talha, Kerem, Kardelen	Whole team
Weekly tracking	Talha	Kardelen	Kerem, Eylül	Whole team

The allocation follows the M1 technical roles. Talha coordinates weekly evidence while Kardelen remains accountable for milestone tracking. Each row has one accountable person; assignments describe future work, not completed implementation.

5.2 Communication Plan

Event / Channel	Purpose	Participants	Frequency	Output
GitHub Issues/Project	Track work and blockers	All	On change; Sunday 20:00 review	Owner, status, next action, evidence
Main commits and issue review	Check substantive changes before Done	Author plus another member	Each substantive change	Reviewed commit and correction note
Team chat	Resolve short questions and urgent blockers	All	As needed; blockers same day	Decision copied to issue/report
Weekly check	Compare plan, actuals and capacity	All	Sunday 20:00 or agreed replacement	Short decision and risk note
Weekly report	Evidence, hours, risks, next work	Talha coordinates; all contribute	Each reporting week	Shared report and DYS receipt if required
Milestone report	Consolidate and	Kardelen	Before published	Final file and DYS

	submit team report	coordinates; all review	DYS deadline	receipt

6. Initial Risk Register

ID	Risk	Probability	Impact	Priority	Mitigation / Response	Owner	Status
R-01	Exam and coursework load	High	Medium	High	Cap work at 5–6 h/member/week; replan weekly.	Kardelen	Open
R-02	Supabase service limits	Medium	High	High	Prototype early and check usage.	Kerem	Open
R-03	Concurrent booking conflict	Medium	High	High	Enforce in data layer; test simultaneous requests.	Kardelen	Open
R-04	Android layout differences	Medium	Medium	Medium	Check two screen sizes.	Talha	Open
R-05	Mobile/API integration delay	Medium	High	High	Agree request/response examples; use sample data.	Kerem	Open
R-06	Scope expansion	Medium	Medium	Medium	Complete Must items; defer Could reminder if needed.	Kardelen	Open
R-07	Missing product issue links	High	Low	Medium	Issues #33–#39 created and backlog linked.	Talha	Resolved for M2
R-08	Estimate uncertainty	High	Medium	High	Record actual hours and reforecast after Sprint 1.	Kerem	Open
R-09	Course dates differ from internal targets	Medium	High	High	Check each DYS assignment; change sprint targets.	Kardelen	Open

7. Baseline Approval

Working baseline adopted for project planning on 1 November 2026 at Kerem's direction: the Must/Should/Could scope in Section 2, 141.5 person-hours and 5–6 h/member/week in Section 3, internal eight-week schedule in Section 4, and RACI in Section 5. Academic M3–M5 deadlines are not yet part of this baseline because they were not verified. The team uses this plan and records changes as work proceeds; adoption does not claim a completed app or test result.

For a scope, effort, schedule or significant technical change, open or update a GitHub issue with the reason, affected items, hour/date impact and alternatives. The affected owner and one other member review the proposed decision. After that review, update the backlog, estimates, schedule, risk register and report as applicable, with a linked `main`

commit. Report material changes in M4. An official DYS date overrides an internal target after the team has recorded its impact.

Definition of Done: the acceptance criterion is demonstrated on a named build or document; relevant tests or document checks are recorded; a teammate reviews the linked `main` commit; documentation and issue status are updated. Planned tests are not counted as passed tests. This rule is shared by the quality plan and team agreement.

8. Planned vs. Actual

Planned activity	Actual result	Deviation	Reason	Corrective action
Backlog and issue links (#6)	Ten work packages; issues #33–#39 created; TSK-08 maps to #18/#21.	Initial draft omitted some M1 features and links.	Backlog was drafted before complete cross-check.	Updated priorities and scope against M1.
Estimates and assumptions (#7)	estimates.xlsx and assumptions.md record 141.5 h and 5–6 h/week capacity.	Earlier report draft counted 86.7 h and omitted support work.	Prototype, UI, integration and test effort were not included.	Use full estimate; track actual hours weekly.
Schedule, RACI and communication (#8)	Source files exist; internal dates and one-A-per-activity RACI prepared.	Official M3–M5 dates not verified.	DYS dates were unavailable at preparation.	Check DYS and update dates on publication.
Quality plan and baseline report (#9)	Quality criteria updated for sign-in, status, reminder, cancellation and search.	No implementation tests have run.	Application development has not started.	Run and record tests on named builds.
M2 DYS submission	Final Word/PDF prepared for the shared team report.	Submission is a separate action.	One member must upload through DYS.	Verify the receipt before 1 Nov 23:59.

9. Next Milestone Plan

Item	Owner	Target date	Expected evidence
Prototype and sign-in increment	Kerem with Talha	8–15 Nov internal target	Named build, setup findings and issues #33/#34
Equipment, booking and search increment	Kerem, Kardelen, Talha	29 Nov internal target	Linked commits and issue evidence #34/#35/#39
Mobile/API integration and acceptance checks	Eylül	27 Dec internal target	Test notes, conflicts and authorization results
M3 feedback, progress chart and report	All; Kardelen coordinates	Official DYS date pending	Feedback notes, actual hours, updated risk register

10. Individual Contributions

Team member	Contribution	Evidence
Talha Yılmaz	Prepared M2 backlog; product issue links #33–#39 now map planned work.	docs/m2/backlog.md; issue #6

Eylül Melda Çakmak	Prepared the M2 quality-plan source; acceptance criteria consolidated in this baseline.	docs/m2/quality-plan.md; issue #9
Kerem Tuna	Directed the capacity-based working allocation and the M2 estimate/assumption files.	docs/m2/estimates.xlsx; docs/m2/assumptions.md; issue #7
Kardelen Darğın	Prepared team agreement and source schedule, RACI and communication materials; coordinates milestone plan.	docs/m2/team-agreement.md; docs/m2/schedule.xlsx; docs/m2/raci.md; docs/m2/communication-plan.md; issue #8

Appendix: Evidence

Repository: https://github.com/keremtuna0/good_tracker

Project board: https://github.com/users/keremtuna0/projects/2

Backlog / WBS: https://github.com/keremtuna0/good_tracker/blob/main/docs/m2/backlog.md

Quality plan and acceptance criteria: https://github.com/keremtuna0/good_tracker/blob/main/docs/m2/quality-plan.md

Team agreement: https://github.com/keremtuna0/good_tracker/blob/main/docs/m2/team-agreement.md

Estimates, assumptions, schedule, RACI, communication and risks: https://github.com/keremtuna0/good_tracker/tree/main/docs/m2

Dated project board snapshot: https://github.com/keremtuna0/good_tracker/blob/main/docs/m2/project-board-snapshot.md